Programming Fundamentals LAB – Spring 2026

(BS-IT-F25 Morning & Afternoon)

LAB # 7

Submission Deadline: Saturday, March 21, 2026 (10:00 PM)

Instructions

- This is an individual lab. Absolutely NO collaboration is allowed. Any traces of plagiarism/cheating or usage of AI tools will result in an "F" grade in this course and lab.
- Do NOT copy even a single line of code from any other person or book or Internet or any other source.
- This assignment needs to be submitted in Soft form. The Submission Procedure is given below.
- Late submissions will NOT be accepted, whatever your excuse may be.

Submission Procedure

1. Make 4 .CPP files for the 4 tasks. Name each file in the following format:

```
TASK1.cpp
TASK2.cpp
TASK3.cpp
TASK4.cpp
```

Include your Name, Roll Number, and Section in comments at the top of each CPP file.

2. Create an empty folder. Place only the .CPP files inside. Do NOT include any other files — your submission will NOT be graded otherwise.
3. Compress the folder as a .ZIP file. Name it exactly as:

```
AI-17-BAIF25M123   (Morning section)
IT-17-BITF25A456   (Afternoon section)

Replace the highlighted part with your complete Roll Number.
```

4. Carefully check your .ZIP file to ensure all instructions above have been followed.
5. Submit the single .ZIP file on Google Classroom. Make sure you press the SUBMIT button after uploading.

Students Without Laptops:

- Write your Name, Roll Number, and Section at the top of each page.
- Scan your handwritten lab in PDF format using CamScanner, Adobe Scan, or similar app.
- Rename the PDF using the same format provided above.
- Submit your PDF file through Google Classroom.

Note: If you do not follow the above submission procedure, your LAB will NOT be graded and you will be awarded ZERO marks.

TASK # 1	Pattern with Skipping Numbers	15 Marks	20 min Time

Pattern printing exercises reinforce your understanding of nested loops and arithmetic progressions. Each row in this task prints a sequence of numbers that increases by a fixed step, making it ideal for practising loop control variables and output formatting.

Task Description

Write a C++ program that takes a starting number and number of rows as input, then prints a pattern where each row contains numbers increasing by 3. The first row starts with the starting number and each subsequent row continues the sequence from where the previous row left off, printing one more number per row.

Requirements

- User Input: Ask for a starting number and number of rows.
- Nested Loop Logic: Outer loop controls rows; inner loop prints numbers for each row.
- Each row i prints i numbers (row 1 prints 1 number, row 2 prints 2 numbers, etc.).
- Numbers increase by 3 across the entire output (not reset per row).

Sample Run

```
Enter starting number: 2
Enter rows: 4

2
5  8
11  14  17
20  23  26  29
```

TASK # 2	Star Pattern	15 Marks	20 min Time

Pattern printing is a classic nested loop exercise. Your task is to print a right-angled triangle made of asterisks (*) that grows from top to bottom, based on a user-provided number of rows.

Requirements

- User Input: Prompt the user to enter the number of rows. Valid range is 3 to 10.
- If the input is outside the valid range, display an appropriate error message.
- Use nested loops: the outer loop controls the row number, the inner loop prints the asterisks.
- Row 1 prints 1 asterisk, row 2 prints 2 asterisks, and so on.

Sample Output — Enter number of rows (3–10): 4

```
*
* *
* * *
```

```
* * * *
```

TASK # 3	File Handling — Orders Analyzer	15 Marks	20 min Time

File handling is a critical skill in real-world programming. In this task you will read structured data from a text file, process it using loops and conditions, and produce a formatted summary report.

Background

You are given a text file named orders.txt that contains records of customer orders in a store. The file stores data in the following format:

- Each order consists of two lines:
  1. First line: Order ID and Order Amount
  2. Second line: Customer Name (may contain spaces)
- The sentinel value -1 indicates the end of data.

Sample Input File (orders.txt)

```
101 1200
Ali Raza
102 450
Sara Khan
103 980
Usman Tariq
104 300
Hina Ali
105 1500
Bilal Ahmed
-1
```

Your Task

Write a C++ program that:

1. Reads data from the file using fin >> for numeric values and getline() for customer names.
2. Determines the total number of orders and the number of high-value orders (Order Amount ≥ 1000).
3. Finds the order with the highest amount and the order with the lowest amount.
4. Displays the results in the format shown below.

Sample Output

```
2 out of 5 orders are high-value orders

Highest order:
  Customer Name: Bilal Ahmed
  Order ID: 105
  Amount: 1500

Lowest order:
  Customer Name: Hina Ali
  Order ID: 104
  Amount: 300
```

TASK # 4	Temperature Log Analyzer	15 Marks	20 min Time

Combining file I/O with statistical analysis is a common real-world programming pattern. In this task you will write temperature data to a file and then read it back to compute useful metrics.

Task Description

A weather station records daily temperatures. Your program should write temperature readings to a file, then analyse the data by reading it back. Follow the steps below:

5. Prompt the user to provide the name of the location being monitored. The location name may consist of multiple words (e.g., "New York", "Lahore City").
6. Ask the user to specify the number of entries to record. The valid range is 3 to 7 entries. For each entry, request a temperature value (decimal values allowed) and persist each value into a file named temperatures.txt, placing each value on a separate line.
7. Once all entries are saved, notify the user with an appropriate confirmation message.
8. Access the saved data from temperatures.txt for further processing. If the file cannot be accessed, display an error and terminate the program gracefully.
9. Process all stored values to identify and display: the lowest recorded value, the highest recorded value, and the average of all recorded values.
10. Additionally, retain all values in memory during the reading phase. Once the average is determined, perform a secondary evaluation to count and display how many entries exceed the calculated average.

Sample Output

```
Enter city name: Lahore City
Enter number of days: 4
Day 1 temperature: 38.5
Day 2 temperature: 41.0
Day 3 temperature: 36.2
Day 4 temperature: 39.8
Data saved to temperatures.txt

--- Temperature Analysis for Lahore City ---
Min Temp   : 36.2 C
Max Temp   : 41.0 C
Average    : 38.9 C
Days above average: 2
```

GOOD LUCK!